



Mission 2: The Sound & Style Master

Objective: Learn to control volume (loudness), use variables to remember numbers, and design your own custom dice patterns.

Phase 6: The Mute Button (Level: Apply)

Goal: Use Button A to "Mute" (silence) and "Unmute" your dice.

- **The Concept:** In coding, silence is just a number.
- **Loud:** `set_volume(128)`
- **Silent:** `set_volume(0)`
- **The Logic:** You need the micro:bit to "toggle." This means checking: "If the volume is 0, make it loud. Otherwise, make it 0."



Lab Check 6: Fill in the Blanks

1. To turn off all sound, I should set the volume to ____.
 2. If I want to check if the volume is currently silent, I use the symbol `==`.
- Example: `___ vol == ___:`

Phase 7: Volume Control (Level: Analyze)

Goal: Use Button A to make it louder (+16) and Button B to make it quieter (-16).

- **The Concept (Variables):** You need a "box" to store the volume level. We call this a **Variable**.
- **The Math:**
 - Press A $\rightarrow v = v + 16$
 - Press B $\rightarrow v = v - 16$
- **The Limits:** Sound cannot go below **0** or above **255**.



Lab Check 7: Fill in the Blanks

1. At the very top of my code, I need to create a variable.

Example: `vol = 128`. This is called **initializing** the variable.

2. To make the sound louder by 16, my code should say:

`vol = vol ____ 16.`

1. If my volume is 255 and I press Button A, the number stays 255 because that is the ____.
 2. After I change the number in the "box," I must tell the hardware to change by using the command:
`set_volume(_____).`
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Phase 8: Creative Design (Level: Create)

Goal: Don't use standard numbers! Design your own "Dice Pattern" for the LED screen.

- **The Concept (Images):** The micro:bit can show more than just numbers. You can draw dots just like a real dice.
- **The Design:** Use `Image()` to define which LEDs are on (9 or #) and which are off (0 or .).

Lab Check 8: Fill in the Blanks

1. In a 5x5 LED grid, the number 0 means the light is __, and the number 9 means the light is at maximum brightness.
2. To show a custom image of the number 1 (a dot in the middle), my code would look like this:

```
dice_one = Image("00000:00000:00_00:00000:00000").
```

Which number is missing in the middle to turn on the center light? Answer: __.

3. Instead of `display.show(x)`, I will now use `if x == 1: display.show(_____)`.
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Final Mission Challenge

Can you combine everything?

- **Shake** to roll.
- **Button A** to make it louder.
- **Button B** to make it quieter.
- **The Screen** shows your custom dots instead of a number.